USER MANUAL

RTMcompare

The toolkit that tells you, with numbers, whether the new master is actually better — and what every streaming platform is going to do to it on the way out.

RELEASE
RTMcompare v5.2.0
macOS arm64 · Windows x64

SIGNED · NOTARIZED
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DOCUMENT
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The toolkit that tells you, with numbers, whether the new master is actually better — and what every streaming platform is going to do to it on the way out.

v5.0.8 · macOS arm64 + Windows x64 · everything runs locally

What's in the box

Three apps. They cover the whole mix → master → ship loop, and they talk to each other through one folder on your disk. No accounts, no cloud, no upload.

APP	WHAT IT DOES	WHO REACHES FOR IT
RTMcompare	A/B compare, single-file QC, album batch, Atmos analysis, streaming preview, master-chain render	Mixing · mastering · label QC
RTMprofile	Drop a corpus of your masters in. Out comes a `.json` that becomes a "sound like Ohad" target curve in RTMcompare's Match tab	Mastering engineers building a personal reference
RTM Send	VST3 / AU / Standalone plugin. One button on your DAW master bus → audio + DAW context lands in RTMcompare	Anyone tired of the bounce-and-drag dance

The whole stack is signed (Apple Developer ID), notarized, and stapled. No Gatekeeper warning. No "this app is from an unidentified developer" speed-bump. Drag, eject, open. Done.

Install

MACOS (ARM64 / APPLE SILICON)

1. Download the DMG.
2. Drag the app onto **/Applications**.
3. Eject the DMG.
4. Launch from /Applications.

That last step matters. macOS App Translocation runs DMG-launched apps inside a sandbox that breaks the bundled Python — so the app will politely refuse to start and tell you to drag it to /Applications first. We'd rather a clean error than a mysterious one.

Three DMGs ship inside the bundle:

- **RTMcompare-5.0.8-arm64.dmg** — the analyser
- **RTMprofile-1.0.5-arm64.dmg** — the profile builder
- **RTM-Send-1.0.0.dmg** — drag the AU/VST3/Standalone bundles into their plugin folders, or double-click `Install RTM Send.command` and let it do the placement for you

WINDOWS (X64)

- **RTMcompare Setup 5.0.8.exe** — NSIS installer (signed)
- **RTMcompare 5.0.8.exe** — portable build (no install, runs in place)
- **RTMprofile Setup 1.0.5.exe** + **RTMprofile 1.0.5.exe** — same pattern

The Windows VST3 is built separately on a Windows host via GitHub Actions. Drop the resulting installer into `Common Files\VST3` and your DAW will find it on next scan.

The five workflows

These are the only five things you'll actually do in this app. Read this section once and you're fluent.

1. TWO-FILE COMPARE — THE MAIN EVENT

Drop **A** (the reference, the rough mix, the previous master) and **B** (the new master, the candidate). Hit **Compare**.

Seven tabs come back at you:

- **Overview** — LUFS / TP / LRA / dynamics deltas, level-matched waveforms, level-matched A/B player. The headline numbers, all in one breath.
- **Mastering** — what your master pass actually changed. Per-band shape, transient density, limiter behaviour, stereo width per band. (The receipts. See *Mastering Delta* below.)
- **Stereo & Spectrum** — vectorscope timeline, per-band phase correlation, mono fold-down per band, masking overlap.
- **Quality** — clip / hum / click / distortion / generation-loss detection. Catches the AAC-of-AAC-of-WAV file the label re-sent you and swore was "the original master."
- **Delivery** — Streaming Preview for every major platform, Apple Digital Masters check, BWF metadata pane, Spec Drift badge.
- **Atmos** — appears when you drop an ADM BWF; channel layout, binaural TP, downmix vectorscope, downmix phase timeline.
- **Match** — the target-curve panel. Pick a reference profile, see the per-band correction, export it as a real DAW preset.

Tabs stick to the top of the scroll. You can bury yourself in a panel and still know where you are.

2. SINGLE-FILE QC

One file, no reference. Click **Analyze Reference Only**.

Use this when a master lands on your desk and you need a clean delivery readout without picking a reference. LUFS, TP, LRA, streaming previews, clip detection, spec compliance — same Delivery tab, faster.

3. ALBUM BATCH

Drop a folder of WAVs / AIFFs / FLACs. Every track gets scanned for LUFS, TP, LRA, ISRC, sample-rate consistency. Outliers vs the album median light up red.

Inside the batch:

- **Cohort Mode** — pin one track as the cohort reference. Every other track gets a heatmap distance score across 31 bands plus an RMS distance column. Sort by distance, find the one track that doesn't sound like family.
- **Loudness anchor** — flip the Δ column from *vs album median* to *vs Spotify −14*, *vs Apple −16*, *vs R128 −23*. Read delivery headroom directly off the table.
- **Reissue mode** — pin "old master" as the A across every song tab. A/B every track against its original.
- **Per-song notes + album notes** — both ride along in the PDF export.

The right-hand song slot rotates as you click rows in the table, so you don't end up with thirty tabs open and a confused desktop.

4. ATMOS / ADM

Drop a multichannel ADM BWF. The analyser auto-routes to the Atmos surface: channel layout, programme name, binaural true-peak, downmix vectorscope, downmix phase correlation timeline. The A/B player auto-loads the Atmos stereo downmix against your stereo reference, level-matched.

Atmos Solo mode handles single-file Atmos QC — same idea, no reference needed.

5. PLUGIN → APP HANDOFF

Install **RTM Send** in your DAW. Drop it on the master bus. Hit the **Send** button.

The plugin writes the rendered audio + DAW context (region name, session name, sample rate, BPM, key, region time range) into `~/.rtm/incoming/`. RTMcompare's file watcher picks it up in about a second and routes it where you asked:

- **Single** → loads as File A, runs analyser-only
- **Compare B** → loads as File B in your active compare
- **Batch** → seeds File A and starts a batch with this track as track 1

A chip appears in the banner with the DAW context and the region timestamp. No bounce dialog. No drag-drop. No remembering where your DAW writes its bounces.

RTMprofile — building your own target reference

Turn ten years of your work into a single `.json` file you can match against.

Open RTMprofile. Drop your back catalogue (five tracks minimum for a stable curve). Type your name, your role, your genres. Click **Build profile**.

The Python side (the same one that ships inside RTMcompare) measures each file:

- LUFS-I via pyloudnorm
- True peak (4× resampled per BS.1770-4)
- LRA, peak-to-RMS, crest
- Third-octave spectrum (31 bands, Welch's method, mean-centred)
- Stereo width

Then it aggregates: median curve, mean+std for the scalars. Out comes a `.json` profile that drops into `~/.rtm/profiles/<slug>.json` and shows up in RTMcompare's Match tab on the next session.

What you do with it:

- Build a *house style* for a label's roster
- Maintain consistency across years of mastering work
- Clone the sound of an engineer you respect (drop their catalogue in, build the profile, match against it)

Privacy note: every measurement runs locally. The profile JSON is anonymised numbers — no audio embedded, no filenames in the output, just a sample count and a curve.

RTM Send — the plugin

JUCE-based. Ships as VST3 + AU + Standalone on macOS, VST3 + Standalone on Windows. (AAX is on the roadmap once we have a PACE Eden cert in hand.)

Three capture modes:

- **Last-N seconds** — a lock-free SPSC ring buffer captures the most recent 30 s (configurable up to 120 s) of audio at the master bus. The default. Always available.
- **Loop region** — uses your DAW's loop points. Plays the loop once, captures it, ships it. Greys out gracefully on hosts that don't expose loop info.
- **Triggered region** — manual Rec / Stop. Host-agnostic. Works in any DAW, including the ones that don't expose loop points.

ARA2 (the "send Track 03 of the Wavelab montage without master-bus playback" feature) is wired up but disabled in the shipped binary. It lights up on the next plugin release.

Three send routes:

- **Single** — writes a `.wav` + `.rtm.json` sidecar + `.ready` marker into `~/.rtm/incoming/`. RTMcompare loads it as the current single-file analysis.
- **Compare (File B)** — same write, different sidecar route hint. Slots into your active two-file compare as B.
- **Album batch** — seeds the batch with this track as track 1.

Rendered audio uses the host's sample rate. The sidecar carries the DAW context (session name, region name, ARA region bounds if relevant) plus a `pluginVersion` stamp so RTMcompare knows what wrote it.

> *Why three apps and not one DAW plugin that does everything? Because the analysis is heavy — Demucs stem separation, 4× true-peak, AAC encoding for the streaming preview, third-octave across 31 bands, the whole Mastering Delta panel — and an audio plugin's host process is the worst possible place to do that work. The plugin is a 1 MB capture-and-handoff agent. The 256 MB analyser sits in its own process where it belongs.*

The headline panels

The seven things that make this not just another meter plug-in.

MASTERING DELTA

The seventh tab on every two-file compare. Treats the comparison as *"what did this master pass actually change?"* and gives you a signed report card:

- Broadband loudness gain (signed, in dB)
- Per-band tonal shape across 31 1/3-octave bands, mean-centred so you're reading the *shape* move, not the *level* move
- LRA delta · PSR delta · RMS-to-peak delta
- Limiter aggressiveness estimate (TP delta-derived)
- Transient density change (%)
- Stereo width change per band
- TP overs pulled back (count)
- **Playback delta after platform normalization** — what the listener actually hears once Spotify / Apple / etc. re-equalize the loudness. On hot masters, often reads 0.0 dB on every platform. That's not a bug — that's streaming normalization wiping your loudness gain in real time. Useful insight.
- **Mastering signature hash** — 8 characters of fingerprint over the rounded per-band shape. Same hash twice = identical move. Catches the "remaster" that wasn't actually re-mastered.

SOUND CHECK TWIN

The teal-green ≈ button next to each platform row in the Streaming Preview. Renders 30 s of the loudest section of your track *through that platform's actual ingest chain*: normalization gain → 4× oversampled true-peak limiter modeled on Apple's, with the per-block gain-reduction envelope visible in the Streaming Delta Heatmap → AAC 256 kbps via Apple's `afconvert` on macOS (ffmpeg fallback on Windows).

Hit ≈. Listen. Decide. The fastest answer to *"will my master clip on Spotify"* you'll ever get.

STREAMING PREVIEW

A live table showing what each platform will *actually play your master at*: normalization action, played LUFS, played TP, breach flags. Spotify, Spotify Loud, Apple Music, YouTube, Tidal, Amazon, Deezer, SoundCloud — plus short-form (TikTok / Reels / Shorts).

Every spec is pinned in a versioned registry — published date, revision date, source citations. Reload an old result a year from now and a **Spec Drift** badge tells you whether the targets you measured against are still current.

MATCH TAB + MASTER ASSISTANT

Pick a reference profile (an engineer's, an album average, your RTMprofile output, or a streaming target). The Master Assistant computes a per-band corrective EQ, optional dynamics, optional limiter — and lets you:

- **Audition it live** through the A/B player with an Amount fader (0–100%) so you can hear the move at any intensity
- **Export the EQ** as a real DAW preset:
- **FabFilter Pro-Q 3 / 4** — native binary `.ffp` (Pro-Q 4 reads Pro-Q 3 binaries unchanged)
- **FabFilter Pro-Q text** — paste into Pro-Q's Paste bar
- **Ableton EQ Eight** — `.adv` (gzipped XML, real Live 12 schema)
- **CSV / JSON / clipboard** — for everything else
- **Apply and bounce** — render a 24-bit WAV with the EQ baked in, optional true-peak limiter at −0.3 dBTP (Apple Music spec, the strictest common ceiling), BWF chunk pre-stamped

TRANSLATION CHECK

Four playback environments rendered as 30 s `.m4a` auditions you can A/B against the original:

- **Phone speaker** — modern phone driver; sub disappears, presence range dominates
- **Earbuds** — consumer earbuds (AirPods-ish)
- **Club PA** — house-system PA with mono-sum sub
- **Car cabin** — generic mid-class consumer car cabin

The renderer tells you the lost-LF energy and the presence-band change before you hit play, so you know what you're walking into.

PER-STEM AI DETECTION (DEEP SCAN)

Demucs separation: vocals / drums / bass / other. Each stem gets its own AI-content verdict. The point is to tell you *where* the suspicion lives — a vocal-cloned-but-organic-instrumentation mix scores very differently from a fully synthetic one, and you want to know which one is on your desk.

REFERENCE LIBRARY

Star a track in the recent-references row. It persists across restarts. The library shows quick-scan stats per reference (LUFS, TP, BPM, key) so you can pick the right reference in three seconds instead of remembering which folder you put it in.

Working modes

The header has a Surface picker. Pick the surface that matches the work; the panels you don't need disappear.

- **Music** — Spotify / Apple Music / YouTube / Tidal / Amazon at top. Broadcast hidden. Atmos hidden.
- **Full** — everything on. Music + broadcast + Atmos.
- **Bcast** — broadcast first: R128 / ATSC A/85 at top, dialog gate prominent.
- **Netflix** — Netflix Sound Mix Specifications v1.6 (−27 LKFS dialog anchor, −2 dBTP ceiling, 5.1 + stereo, 48k/24-bit minimum).
- **Post** — Atmos / immersive. ADM validation surfaced.

Two side toggles:

- **Advanced QC** — reveals collapsed-by-default diagnostic panels (Mono Compatibility, Phase Bands, Transient Density, Tempo Drift, Waveform Diff, Masking Overlap). Off by default because most engineers don't need them most of the time. Needs Deep Scan to populate.
- **Educator** — adds a *why this matters* explainer to every panel. Useful when you're handing the report to a producer or onboarding someone junior.

Keyboard shortcuts

Press `?` anywhere for the full list. The ones you'll actually use:

KEY	ACTION
`Space`	A/B player play / pause
`Tab`	toggle A / B
`M`	mono fold-down on the player
`1` – `7`	jump to tabs 1–7 in compare view
`[` / `]`	set loop start / loop end
`\`	clear loop
`←` / `→`	step between songs in batch song-tab
`⌘/Ctrl + K`	command palette — search any metric, jump to its tab
`⌘/Ctrl + E`	export EQ
`⌘/Ctrl + ⇧ E`	apply EQ + bounce

Where your data lives

Everything stays on your disk. Always.

- **macOS**: `~/Library/Application Support/RTMcompare/python-cache/` — Python bytecode + Numba JIT cache, redirected here so the signed app bundle is never written to (codesign integrity stays intact)
- `~/.rtm/` — analysis history, reference library, plugin inbox, custom DSP profiles. Lives in your home folder so it survives app reinstalls
- `~/.rtm/profiles/` — engineer profile JSONs (RTMprofile output lands here)

No telemetry. No phone-home. No analytics. The only network calls happen when you sign in with a license — and even then the app works fully offline for 180 days after activation.

Troubleshooting

The five things people actually run into.

"RTMcompare cannot be opened from inside the DMG" You launched from the DMG window. macOS App Translocation runs DMG-launched apps in a sandbox that can't reach our bundled Python. Drag to /Applications, eject the DMG, open from there.

"Preparing audio..." stuck on the player Pre-5.0 bug. Update to 5.0.5+ (you're already on 5.0.8 if you're reading this).

Sound Check twin shows "render ×" on Windows Install ffmpeg (`winget install ffmpeg`) and add it to PATH. Restart RTMcompare.

Advanced QC panels are blank You're viewing a Fast scan result. Re-run with Deep Scan.

Mastering Delta says 0.0 dB on every platform Working as designed. Both files are louder than every platform's normalization target, so streaming normalization attenuates them to the same level. The metric is correctly telling you that mastering loudness gains evaporate after platform normalization. This is the lesson, not the bug.

Plugin → App handoff doesn't show up in RTMcompare Check `~/.rtm/incoming/` exists and the plugin can write to it. RTMcompare must be running for the file watcher to fire. The plugin writes `.ready` last; if you see `.wav` + `.rtm.json` but no `.ready` , the plugin crashed mid-write — check the host's plugin log.

RTMprofile build fails with "Python not found" RTMprofile probes for RTMcompare's bundled Python first. If RTMcompare isn't installed, it falls back to system `python3` — which needs `numpy` , `scipy` , `soundfile` , `pyloudnorm` already installed. Install RTMcompare and the problem disappears.

Versions + spec dates

The app shows its version in the lower-right of the header. Each analysis stamps the spec versions it ran against. Reload an old result and the **Spec Drift** badge tells you if anything has shifted since.

This manual covers **5.0.8**.